TEXT ANALYSIS USING PYTHON

1. Introduction

Text Analysis is a Python-based project that processes a given paragraph and calculates useful textual statistics. The program determines the number of words, characters, vowels, and repeated words.

Text analysis is a basic technique used in Natural Language Processing (NLP), where computers process and analyze human language.

2. Objective

The objectives of this project are:

- To count the words in a given text.
- To calculate the number of characters.
- To count occurrences of vowels.
- To identify repeated words and their frequencies.
- To practise basic text-processing techniques using Python.

3. Problem Statement

Text contains useful information, but manually counting words, characters, vowels, and repeated terms can be time-consuming.

This project automates these calculations using Python and presents the results in a structured format.

4. Tools and Technologies Used

Python 3: Used to implement text processing and statistical calculations.

Regular Expressions: Used to identify words while handling punctuation.

Counter: A Python utility used to calculate word frequencies.

Visual Studio Code: Used to develop and run the program.

GitHub: Used to store the source code and documentation.

5. Methodology

The project follows these steps:

1. Obtain sample text or accept text entered by the user.
2. Convert text to lowercase for case-insensitive word counting.
3. Extract words using regular expressions.
4. Calculate the word count.

5. Calculate the character count using the original text.
6. Count occurrences of the vowels a, e, i, o, and u.
7. Calculate the frequency of each extracted word.
8. Identify words occurring more than once.
9. Display the analysis results.

6. Features

Word Count

Counts the words extracted from the input text. Punctuation is excluded from word tokens.

Character Count

Counts every character in the original text, including spaces, punctuation, and digits.

Vowel Count

Counts the occurrences of the five vowels: a, e, i, o, and u. Uppercase and lowercase vowels are treated equally.

Repeated-Word Analysis

Identifies words that occur more than once and displays their frequencies.

Unique-Word Count

Calculates the number of distinct word tokens in the text, ignoring capitalization.

7. Sample Dataset

The program includes the following sample paragraph:

"Python is useful for text analysis. Python can count words and characters. Text analysis helps us understand words."

The program can also accept custom text from the user.

8. Results

The program displays:

- Total word count.
- Total character count.
- Total vowel count.
- Number of unique words.
- Repeated words and their frequencies.

For the supplied sample paragraph, the expected word count is 18.

Other results should be verified by running the program.

Screenshot Evidence: Insert a screenshot of the actual program output here.

9. Testing and Validation

The program should be tested with:

- A normal sentence.
- Repeated words.
- Uppercase and lowercase text.
- Text containing punctuation.
- An empty string.
- Text containing digits.
- Text without vowels.
- Text containing multiple spaces.

Actual test outcomes should be recorded and compared with expected results.

10. Concepts Learned

The project demonstrates:

- Python strings.
- String processing.
- Regular expressions.
- Loops and conditional statements.
- Dictionaries and frequency counting.
- Python's `Counter`.
- Basic Natural Language Processing concepts.
- Testing and validation.

11. Limitations and Future Improvements

The current program performs basic text analysis and does not interpret the meaning or sentiment of text.

Future improvements could include:

- Reading text from files.
- Generating word-frequency charts.
- Removing common stop words.
- Performing sentiment analysis.
- Extracting keywords.
- Building an interactive interface.

12. Conclusion

The Text Analysis project demonstrates how Python can transform textual information into numerical statistics. It calculates word and character counts, counts vowels, and identifies repeated words.

The project provides practical experience in Python programming and introduces techniques that can be extended to more advanced Natural Language Processing applications.

13. Project Links

GitHub Repository: https://github.com/venky1318/text-analysis